

6

NUMBER AND ALGEBRA

EQUATIONS, INEQUALITIES AND LOGARITHMS

In 1962, when astronaut John Glenn was preparing to be the first American to orbit the Earth, he called upon mathematician Katherine Johnson (1918–2020) to check by hand the complex calculations being performed on NASA's new electronic computers.
Johnson worked for NASA for 35 years and her story was told in the 2016 book and film *Hidden Figures*. During her career, she solved equations that guided the paths of spacecrafts and helped the US send the first astronauts to the moon in 1969. Building on the ancient achievements of the Egyptians and Babylonians, Johnson opened the door to space travel and future space tourism.



Chapter outline

		Working mathematically				
6.01	Equations	U	F			
6.02	Equations with algebraic fractions*	U	F			
6.03	Quadratic equations $x^2 + bx + c = 0$*	U	F		R	C
6.04	Simple cubic equations $ax^3 = c$*	U	F		R	C
6.05	Equation problems	U	F	PS	R	C
6.06	Equations and formulas	U	F		R	C
6.07	Changing the subject of a formula*	U	F		R	C
6.08	Graphing inequalities on a number line*	U	F			C
6.09	Solving inequalities*	U	F		R	C
6.10	Logarithms*	U	F		R	C
6.11	Logarithm laws*	U	F		R	C
6.12	Logarithmic scales*	U	F		R	C
6.13	Exponential and logarithmic equations*	U	F	PS	R	

***PATH**

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Wordbank

> 'is greater than'

< 'is less than'

cubic equation An equation involving a variable cubed (power of 3), such as $4x^3 = 500$

exponential equation An equation where the variable is a power, such as $3^x = 243$

Quiz
Wordbank 6

inequality A mathematical statement that 2 quantities are not equal, involving algebraic expressions and an inequality sign (>, ≥, <, or ≤)

logarithm The power of a number, to a given base. For example, $\log_{10} 1000 = 3$, meaning that the logarithm of 1000 to base 10 is 3, because $1000 = 10^3$

quadratic equation An equation involving a variable squared (power of 2), such as $3x^2 - 6 = 69$

solution The answer to an equation, inequality or problem; the correct value(s) of the variable that makes an equation or inequality true

Videos (17):

6.01 Equations with variables on both sides • Equations with brackets
6.02 Equations with algebraic fractions • Solving equations with fractions
6.03 Simple quadratic equations • Quadratic equations by factorising
6.06 Equations and formulas 2 • Equations and formulas 1 • Formulas and equations
6.07 Changing the subject of a formula
6.08 Graphing inequalities on the number line
6.09 Solving inequalities
6.10 Logarithms
6.11 Logarithm laws
6.12 Logarithmic scales
6.13 Exponential equations 2 • Exponential equations 1

Twig video (1):

6.12 The Richter scale

Quizzes (5):

- Wordbank 6
- SkillCheck 6
- Mental skills 6
- Language of maths 6
- Test yourself 6

Worksheets (5):

6.07 Changing the subject of a formula
6.08 Graphing inequalities
6.09 Inequalities review
6.13 Logarithms review
Mind map Equations, inequalities and logarithms

Puzzles (12):

6.01 Equations with unknowns on both sides • Equations
6.02 Equations code puzzle • Equations order activity • Solving linear equations 1 • Solving linear equations 2
6.06 Getting the right formula
6.07 Formulas squaresaw
6.10 Logarithms 1 • Logarithms 2
6.13 Exponential equations
Language of maths Equations and inequalities crossword

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- solve linear equations and problems involving equations
- (PATH) solve linear equations involving algebraic fractions
- solve quadratic equations of the form $ax^2 = c$ and (PATH) $x^2 + bx + c = 0$#
- (PATH) solve simple cubic equations of the form $ax^3 = c$
- use formulas to solve problems
- (PATH) change the subject of a formula
- (PATH) graph inequalities on a number line
- (PATH) solve linear inequalities
- (PATH) apply logarithms and the logarithm laws and use logarithmic scales
- (PATH) solve exponential and logarithmic equations

#Note: More coverage of quadratic equations in Chapter 10, *Quadratic equations and the parabola*.

SkillCheck ANSWERS ON P. 630

1 Solve each equation.

a $4a + 5 = 2a - 19$ **b** $\frac{3x+2}{5} = 4$ **c** $4(2 - x) = -24$

2 Factorise each expression.

a $k^2 + 5k + 4$ **b** $y^2 - 10y + 16$ **c** $m^2 - m - 56$

d $u^2 + 8u - 65$ **e** $w^2 - 10w + 21$ **f** $x^2 - 2x - 24$

Quiz
SkillCheck 6

Equations 6.01

Example 1

Solve each equation.

a $4(y + 1) + 3(y - 5) = 8$ **b** $\frac{6-m}{5} = 4$

Solution

a Expand and collect like terms.

$4(y + 1) + 3(y - 5) = 8$

$4y + 4 + 3y - 15 = 8$

$7y - 11 = 8$

$7y - 11 + 11 = 8 + 11$

$7y = 19$

$\frac{7y}{7} = \frac{19}{7}$

$y = 2\frac{5}{7}$

Check:

$\text{LHS} = 4\left(2\frac{5}{7} + 1\right) + 3\left(2\frac{5}{7} - 5\right) = 8$

RHS = 8

LHS = RHS

b

$\frac{6-m}{5} = 4$

$\frac{6-m}{5} \times 5 = 4 \times 5$

$6 - m = 20$

$6 - m - 6 = 20 - 6$

$-m = 14$

$m = -14$

Puzzles
Equations with unknowns on both sides

Equations

Videos
Equations with variables on both sides

Equations with brackets

EXERCISE 6.01 ANSWERS ON P. 630

Equations U F

1 Solve each equation.

a $5y + 10 = 3y + 30$ **b** $8a + 20 = 4a + 10$ **c** $6y - 1 = 3y + 14$

d $12a + 30 = 5a + 9$ **e** $5y + 3 = 8y - 21$ **f** $14x - 20 = 8x - 14$

g $9y + 1 = 3y - 5$ **h** $15x - 15 = 8x - 85$ **i** $8m - 10 = 5 - 2m$

j $18 - 3y = 6 - 2y$ **k** $1 - 7a = 10 + 2a$ **l** $11 - 5x = 3x + 43$

2 Solve $4y = y - 15$. Select **A**, **B**, **C** or **D**.

A $y = -3$ **B** $y = \frac{7}{4}$ **C** $y = -5$ **D** $y = 11$

3 Solve each equation.

a $3(x - 6) = 30$ **b** $5(m + 10) = 80$ **c** $2(5y + 3) = 46$

d $3(y + 2) = 5y - 10$ **e** $5(y + 4) = 3y + 6$ **f** $10(x - 3) = 5(x + 5)$

g $2(3m + 6) = 4(m - 1)$ **h** $5(2a + 7) = 5(4 - a)$ **i** $3(1 - 2y) = 18 - 3y$

4 Solve $2(y - 3) = 5 + 4y$. Select **A**, **B**, **C** or **D**.

A $y = -9$ **B** $y = -5$ **C** $y = -\frac{11}{2}$ **D** $y = -\frac{1}{2}$

EXAMPLE 1

5 Solve each equation.

a $3(d + 3) + 4(d + 1) = 15$ **b** $3(y - 1) + 5(y + 4) = 10$

c $7(k + 1) + 2(k - 6) = 3$ **d** $5(g - 3) + 2(g - 2) = 4$

e $6(2h + 3) + 5(h - 3) = 9$ **f** $2(1 + p) + 3(4 + p) = 5$

6 Solve each equation.

a $\frac{3m}{4} + 6 = 10$ **b** $\frac{7 + a}{8} = 3$ **c** $3 + \frac{2d}{7} = 9$

d $\frac{9 - y}{5} = 3$ **e** $8 - \frac{4x}{3} = 5$ **f** $10 = \frac{9b}{7}$

g $\frac{x + 1}{4} + 2 = 10$ **h** $\frac{y - 2}{5} - 6 = 3$ **i** $\frac{m + 2}{5} - 1 = 3$

j $\frac{4h - 3}{2} = 7$ **k** $\frac{2e + 11}{4} = 6$ **l** $\frac{12 - 5y}{3} = 9$

m $\frac{8h + 3}{5} = 7$ **n** $\frac{9a}{4} - 5 = 7$ **o** $\frac{9 - 3c}{7} = 8$

Equations with algebraic fractions 6.02

Example 2

Solve each equation.

a $\frac{5k+4}{3}-7=12$ **b** $\frac{3(p-3)}{2}+6=1$

Solution

a

$$\frac{5k+4}{3}-7=12$$

$$\frac{5k+4}{3}-7+7=12+7$$

$$\frac{5k+4}{3}=19$$

$$\frac{5k+4}{3}\times 3=19\times 3$$ Multiply both sides by 3

$$5k+4=57$$

$$5k+4-4=57-4$$

$$5k=53$$

$$\frac{5k}{5}=\frac{53}{5}$$

$$k=10\frac{3}{5}$$

b

$$\frac{3(p-3)}{2}+6-6=1-6$$

$$\frac{3(p-3)}{2}=-5$$

$$\frac{3(p-3)}{2}\times 2=-5\times 2$$ Multiply both sides by 2

$$3(p-3)=-10$$

$$3p-9=-10$$ Expand the LHS

$$3p-9+9=-10+9$$

$$3p=-1$$

$$\frac{3p}{3}=\frac{-1}{3}$$

$$p=-\frac{1}{3}$$

For equations with more than one fraction, multiply both sides by a common multiple of the denominators to remove the fractions.

Example 3

Solve each equation.

a $\frac{2m}{3}-\frac{m}{2}=2$ **b** $\frac{2a+4}{5}=\frac{2}{3}$ **c** $\frac{3x-5}{4}=\frac{x+1}{7}$

Solution

a

$$\frac{2m}{3}-\frac{m}{2}=2$$

$$6\left(\frac{2m}{3}-\frac{m}{2}\right)=6\times 2$$ Multiply both sides by 6, the lowest common multiple (LCM) of 3 and 2

$$\cancel{6}^{2}\left(\frac{2m}{\cancel{3}_{1}}\right)-\cancel{6}^{3}\left(\frac{m}{\cancel{2}_{1}}\right)=12$$

$$4m-3m=12$$

$$m=12$$

Check by substituting that this solution is correct.

PATH

Puzzles
Equations code puzzle

Equations order activity

Solving linear equations 1

Solving linear equations 2

Videos
Equations with algebraic fractions

Solving equations with fractions

PATH

Note: We can also solve this equation by simplifying the LHS first:

$$\frac{2m}{3}-\frac{m}{2}=2$$
$$\frac{4m}{6}-\frac{3m}{6}=2$$
$$\frac{m}{6}=2$$
$$m=12$$

However, the first method may be faster and easier because operations with fractions are not required.

b

$$\frac{2a+4}{5}=\frac{2}{3}$$
$$15\left(\frac{2a+4}{5_1}\right)=15^{5}\left(\frac{2}{3_1}\right) \quad \text{LCM} = 15$$
$$3(2a+4)=10$$
$$6a+12=10$$
$$6a=-2$$
$$a=\frac{-2}{6}$$
$$a=-\frac{1}{3}$$

c

$$\frac{3x-5}{4}=\frac{x+1}{7}$$
$$28^{7}\left(\frac{3x-5}{4_1}\right)=28^{4}\left(\frac{x+1}{7_1}\right) \quad \text{LCM} = 28$$
$$7(3x-5)=4(x+1)$$
$$21x-35=4x+4$$
$$17x=39$$
$$x=\frac{39}{17}$$
$$x=2\frac{5}{17}$$

Video
Equations with algebraic fractions

Example 4

Solve $\frac{2n+1}{3}-\frac{3n-2}{2}=-5$.

Solution

$$\frac{2n+1}{3}-\frac{3n-2}{2}=-5$$
$$6^{2}\left(\frac{2n+1}{3_1}\right)-6^{3}\left(\frac{3n-2}{2_1}\right)=6\times(-5)$$
$$2(2n+1)-3(3n-2)=-30$$
$$4n+2-9n+6=-30$$
$$-5n+8=-30$$
$$-5n=-38$$
$$n=\frac{-38}{-5}$$
$$=7\frac{3}{5}$$

EXERCISE 6.02 ANSWERS ON P. 630

Equations with algebraic fractions U F

1 Solve $\frac{x-1}{4}+2=10$. Select the correct answer **A**, **B**, **C** or **D**.

A $x=49$ **B** $x=37$ **C** $x=33$ **D** $x=3$

Foundation Standard Complex

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2 Solve each equation.

a $\frac{2y-3}{5}+2=7$ **b** $\frac{7m+5}{2}-3=8$ **c** $\frac{4+3x}{4}-5=2$ **d** $\frac{7+2k}{6}+3=12$

e $6+\frac{3w+1}{4}=10$ **f** $4+\frac{6a-5}{3}=9$ **g** $\frac{1-4e}{2}-8=3$ **h** $6+\frac{3-2x}{3}=0$

i $\frac{5-4p}{6}+2=5$ **j** $\frac{7-3y}{4}+1=5$ **k** $\frac{2(x+1)}{5}=10$ **l** $\frac{3(m-2)}{4}=6$

m $\frac{8(n+1)}{3}+2=4$ **n** $\frac{5(1-n)}{2}-1=3$ **o** $\frac{4(1+d)}{3}+1=7\frac{1}{3}$ **p** $8+\frac{3(w-4)}{5}=14$

q $\frac{4(d-5)}{7}-3=4$ **r** $4=\frac{6(3-m)}{5}-4$ **s** $1+\frac{4(7-h)}{3}=13$ **t** $5=2+\frac{3(1-2g)}{7}$

3 Solve each equation.

a $\frac{2k}{3}=\frac{5}{4}$ **b** $\frac{3w}{10}=\frac{2}{5}$ **c** $\frac{5x}{2}=-\frac{10}{3}$ **d** $\frac{5k-1}{4}=\frac{3}{5}$

e $\frac{5}{4}=\frac{4m+1}{7}$ **f** $\frac{1-2y}{5}=\frac{8}{3}$ **g** $\frac{n-7}{6}=\frac{n+3}{4}$ **h** $\frac{y+2}{5}=\frac{y-1}{2}$

i $\frac{a+5}{3}=\frac{a-1}{8}$ **j** $\frac{p+2}{5}=\frac{p-5}{2}$ **k** $\frac{4k-3}{5}=\frac{2k+7}{4}$ **l** $\frac{3y+2}{3}=\frac{2y+1}{4}$

m $\frac{w}{5}+\frac{w}{2}=7$ **n** $\frac{5a}{7}-\frac{2a}{3}=2$ **o** $\frac{2e}{3}+\frac{e}{2}=8$ **p** $\frac{3a}{2}+\frac{4a}{5}=1$

q $\frac{2y}{5}-\frac{y}{3}=4$ **r** $\frac{a}{3}+\frac{3a}{4}=2$ **s** $\frac{5m}{4}-\frac{3m}{10}=1$ **t** $\frac{4h}{3}+\frac{h}{5}=3$

4 Solve each equation.

a $\frac{x-1}{4}+\frac{2x}{7}=0$ **b** $\frac{p+2}{3}+\frac{p+1}{4}=10$ **c** $\frac{m-2}{4}+\frac{m+1}{6}=12$

d $\frac{x-3}{5}+\frac{x-2}{2}=6$ **e** $\frac{3x-10}{3}+\frac{x-2}{2}=11$ **f** $\frac{3y+1}{4}-\frac{y+2}{3}=4$

g $\frac{7+2a}{5}-\frac{a-1}{2}=6$ **h** $\frac{6a-1}{4}-\frac{a+2}{3}=8$ **i** $\frac{a-10}{5}-\frac{5-2a}{4}=\frac{1}{2}$

PATH

EXAMPLE 2

EXAMPLE 3

EXAMPLE 4

6.03

Quadratic equations $x^2 + bx + c = 0$ 6.03

The quadratic equation $x^2 = c$

The quadratic equation $x^2 = c$ (where c is a positive number) has 2 solutions:

$x = \pm\sqrt{c}$ (which means $x = \sqrt{c}$ and $x = -\sqrt{c}$).

Foundation Standard Complex

Video
Simple quadratic equations

Example 5

Solve each quadratic equation.

a $3x^2 = 75$ **b** $4m^2 - 12 = 0$

Solution

a $3x^2 = 75$

$x^2 = \frac{75}{3}$

$x^2 = 25$

$x = \pm\sqrt{25}$

$= \pm 5$

b $4m^2 - 12 = 0$

$4m^2 - 12 + 12 = 0 + 12$

$4m^2 = 12$

$m^2 = \frac{12}{4}$

$m^2 = 3$

$m = \pm\sqrt{3}$

Leave the answer as a surd (in exact form)

PATH

The quadratic equation $x^2 + bx + c = 0$

To solve quadratic equations of the form $x^2 + bx + c = 0$, factorise $x^2 + bx + c$ into binomial products.

Video
Quadratic equations by factorising

Example 6

Solve $x^2 + 5x + 6 = 0$.

Solution

$x^2 + 5x + 6 = 0$

$(x + 2)(x + 3) = 0$

$\therefore x + 2 = 0$ or $x + 3 = 0$

$\therefore x = -2$ or $x = -3$

$\therefore$ The solution to $x^2 + 5x + 6 = 0$ is $x = -2$ or $x = -3$.

Check:

When $x = -2$,

LHS $= (-2)^2 + 5 \times (-2) + 6 = 0$

RHS $= 0$

LHS = RHS.

When $x = -3$,

LHS $= (-3)^2 + 5 \times (-3) + 6 = 0$

RHS $= 0$

LHS = RHS.

Example 7

Solve each quadratic equation.

a $u^2 + 3u - 28 = 0$ **b** $3m^2 = 6m$ **c** $w^2 - 10w + 25 = 0$

Solution

a $u^2 + 3u - 28 = 0$

$(u + 7)(u - 4) = 0$

$\therefore u = -7$ or $u = 4$

b $3m^2 = 6m$

$3m^2 - 6m = 0$

This requires a simpler factorisation as there are only 2 terms, both involving m.

$3m(m - 2) = 0$

$\therefore 3m = 0$ or $m - 2 = 0$

$\therefore m = 0$ or $m = 2$

You cannot divide both sides by m because if $m = 0$, then you are dividing by 0.

c $w^2 - 10w + 25 = 0$

$(w - 5)(w - 5) = 0$

$\therefore w = 5$ or $w = 5$

$\therefore$ The solution is $w = 5$ (only one solution).

PATH

Video
Quadratic equations by factorising

6.03

Note: Quadratic equations of the form $ax^2 + bx + c = 0$ can be found in Chapter 10, *Quadratic equations and the parabola*.

EXERCISE 6.03 ANSWERS ON P. 631

Quadratic equations $x^2 + bx + c = 0$ UFRC

EXAMPLE 5

1 Solve each quadratic equation, writing the solutions as surds if necessary.

a $\frac{a^2}{2} = 8$ **b** $4k^2 = 180$ **c** $3w^2 = 300$ **d** $d^2 + 60 = 204$

e $\frac{k^2}{2} = 7$ **f** $\frac{w^2}{10} = 2.5$ **g** $4x^2 = 1$ **h** $\frac{m^2}{4} = 10$

i $5y^2 = 5$ **j** $2p^2 + 3 = 21$ **k** $\frac{3k^2}{2} + 1 = 13$ **l** $\frac{y^2}{5} - 2 = 18$

2 Solve each equation, correct to one decimal place where necessary.

a $9m^2 - 2 = 32$ **b** $9k^2 + 10 = 13$ **c** $\frac{2x^2}{5} = 23$

d $\frac{k^2}{16} = 6$ **e** $\frac{3k^2}{10} = 27$ **f** $6y^2 = 0.726$

g $3a^2 + 11 = 267$ **h** $2y^2 - 14 = 63$ **i** $\frac{2w^2}{5} - 1 = 19$

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PATH

EXAMPLE 6

EXAMPLE 7

3 Solve each equation.

a $x^2 + 3x + 2 = 0$ **b** $y^2 + 5y + 4 = 0$ **c** $y^2 + 16y + 48 = 0$

d $x^2 + x - 12 = 0$ **e** $x^2 + 2x - 3 = 0$ **f** $x^2 + 3x - 40 = 0$

4 Solve each equation.

a $x^2 - x - 30 = 0$ **b** $x^2 - 8x + 16 = 0$ **c** $x^2 - 5x - 66 = 0$

d $d^2 - 2d = 0$ **e** $x^2 - 3x - 10 = 0$ **f** $n^2 + 4n = 0$

g $k^2 - 7k = 0$ **h** $y^2 = 5y$ **i** $v^2 = 12v$

5 Explain why the quadratic equation $x^2 = -25$ has no solutions. R C

6 State which of these quadratic equations have no solutions. Give reasons. R C

a $x^2 = -9$ **b** $2k^2 + 5 = 9$ **c** $3m^2 + 8 = 4$

d $\frac{9w^2}{2} - 1 = 1$ **e** $4 + \frac{d^2}{3} = 8$ **f** $\frac{5a^2}{2} + 3 = 2$

7 Solve each quadratic equation. Select **A**, **B**, **C** or **D**.

a $x^2 + 4x - 60 = 0$

A $x = -10, 6$ **B** $x = 12, -5$ **C** $x = 10, -6$ **D** $x = -12, 5$

b $q^2 + 3q = 0$

A $q = 3, -3$ **B** $q = 6, -3$ **C** $q = 0, -3$ **D** $q = 0, 3$

6.04 Simple cubic equations $ax^3 = c$

PATH

The cubic equation $x^3 = c$

The cubic equation $x^3 = c$ has one solution:

$$x = \sqrt[3]{c}.$$

Example 8

Solve each cubic equation.

a $p^3 = 50$ **b** $-2x^3 = 2000$

Solution

a $p^3 = 50$

$p = \sqrt[3]{50}$

50 is not a cube number so leave the answer as a surd

b $-2x^3 = 2000$

$x^3 = \frac{2000}{-2}$

$= -1000$

$x = \sqrt[3]{-1000}$

$= -10$

Foundation Standard Complex

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EXERCISE 6.04 ANSWERS ON P. 631

PATH

Simple cubic equations $ax^3 = c$ U F R C

EXAMPLE 8

1 Solve each cubic equation, writing the solution in exact form where necessary.

a	$x^3 = 1$	**b**	$m^3 = 125$	**c**	$a^3 = 1331$
d	$u^3 = -8$	**e**	$y^3 = -729$	**f**	$n^3 = 20$
g	$h^3 = 11$	**h**	$k^3 = -48$	**i**	$5m^3 = -75$
j	$7m^3 = 448$	**k**	$-4x^3 = 81$	**l**	$12x^3 = -480$

2 Solve each cubic equation, writing the solution correct to one decimal place.

a	$w^3 - 16 = 0$	**b**	$m^3 + 6 = 22$	**c**	$5m^3 - 1080 = 0$
d	$3t^3 - 10 = 87$	**e**	$\frac{x^3}{3} = 9$	**f**	$\frac{5x^3}{7} = -120$
g	$\frac{3x^3}{4} = -10$	**h**	$\frac{2x^3}{5} = 0.2048$	**i**	$\frac{7a^3}{9} - 10 = 121$
j	$a^3 - 0.064 = 0$	**k**	$12 - \frac{7x^3}{8} = 52$	**l**	$5t^3 + 46 = -370$

3 **a** Does a cubic equation of the form $ax^3 = c$ always have a solution? R C

b When is the solution to $x^3 = c$ positive?

c When is the solution to $x^3 = c$ negative?

d Can $x^3 = c$ have 2 solutions?

6.05

Equation problems 6.05

Example 9

At a concert, an adult's ticket costs \$5 more than twice the cost of a child's ticket. The total cost for 3 adults and 7 children is \$327. Find the cost of a child's ticket and an adult's ticket.

Solution

Let the cost of a child's ticket be \$$c$.

Using a variable to represent an unknown quantity

$\therefore$ Cost of an adult's ticket $= \$(2c + 5)$

$3(2c + 5) + 7c = 327$ — Forming an equation

$6c + 15 + 7c = 327$ — Solving the equation

$13c + 15 = 327$

$13c = 312$

$c = 24$ — A child's ticket costs \$24

$\therefore$ Cost of an adult's ticket $= 2 \times \$24 + 5$ — $\$(2c + 5)$

$= \$53$

$\therefore$ A child's ticket costs \$24 and an adult's ticket costs \$53.

Check: $3 \times \$53 + 7 \times \$24 = \$327$

Foundation Standard Complex

Example 10

Paris is 7 years older than Amy. 10 years from now, the sum of their ages will be 43.

How old are they now?

Solution

Let x = Amy's age now.

Then Paris' age now $= x + 7$.

Break the information into 'Now' and 'In 10 years time'.

	Now	In 10 years time
Amy	x	$x + 10$
Paris	$x + 7$	$x + 7 + 10 = x + 17$

In 10 years time:

Sum of ages: $(x + 10) + (x + 17) = 43$

$$2x + 27 = 43$$
$$2x = 16$$
$$x = 8$$

Amy is 8 years old now.

Paris is $8 + 7 = 15$ years old now.

Check: In 10 years time, the sum of their ages will be $18 + 25 = 43$.

EXERCISE 6.05 ANSWERS ON P. 631

Equation problems U F PS R C

For each question, write an equation and solve it to answer the problem. PS R C

1. The longer sides of an isosceles triangle are twice as long as the shorter side. The perimeter of the triangle is 90 mm. Find the lengths of the sides of the triangle.

EXAMPLE 9

2. At the football match, an adult's ticket costs \$6 more than twice the cost of a child's ticket. The total cost for 3 adults and 5 children is \$249. Find the cost of a child's ticket and an adult's ticket.

3. The length of a rectangle is 3 times as long as its width. The perimeter of the rectangle is 152 mm. Find its dimensions.

4. The length of a rectangle is 3 more than twice its width. Find the dimensions of the rectangle if its perimeter is 84 cm.

5. The sum of 3 **consecutive** integers is 186. Find the integers.

EXAMPLE 10

6. Tyson is 9 times the age of his daughter, Charlotte. In 5 years, he will be 4 times the age of Charlotte. How old are they now?

7. When 15 is subtracted from 3 times a certain number, the answer is 63. What is the number?

8. The product of 2 and a number is the same as 12 minus the number. Find the number.

9. The sum of the present ages of Vatha and Chris is 36. In 4 years time, the sum of their ages will equal twice Vatha's present age. How old are they now?

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10 4 consecutive integers have a sum of 858. Find the 4 integers.

11 The sum of 3 consecutive even numbers is 288. Find the numbers.

12 Manori's bag contained 10-cent and 20-cent coins. She had 202 coins, with a total value of $31.90. How many 20-cent coins did Manori have?

13 If 17 more than a number is 5 more than 3 times the number, what is the number?

14 The sum of Scott's age and his mother Kait's age is 45. In 5 years time, 3 times Scott's age less 9 will be the same as Kait's age. Find the present ages of Scott and Kait.

15 One angle in a triangle is double the smallest angle, and the third angle in the triangle is 5 more than 4 times the smallest angle. Find the size of each angle in the triangle.

16 A large container of water is $\frac{7}{8}$ full. After 15 L has been taken out, the container is $\frac{2}{3}$ full. When full, how many litres does the container hold?

17 The total cost of a school camp for Year 10 students was $21 280. Each teacher paid $185 to attend and each student paid $165. There was one teacher for every 15 students. Find the numbers of teachers and students that attended the camp.

Mental skills 6: Maths without calculators ANSWERS ON P. 631

Quiz
Mental skills 6

Multiplying and dividing by 5, 15, 25 and 50

1 Study each example.

a **To multiply by 5**, halve the number, then multiply by 10.

$18 \times 5 = 9 \times 10$

$= 90$

b **To multiply by 50**, halve the number, then multiply by 100.

$26 \times 50 = 13 \times 100$

$= 1300$

c **To multiply by 25**, quarter the number, then multiply by 100.

$44 \times 25 = 11 \times 100$

$= 1100$

d **To multiply by 15**, halve the number, then multiply by 30.

$8 \times 15 = 4 \times 30$

$= 120$

e **To divide by 5**, divide by 10 and double the answer. We do this because there are 2 5s in every 10.

$140 \div 5 = 14 \times 2$

$= 28$

f **To divide by 50**, divide by 100 and double the answer. This is because there are 2 50s in every 100.

$400 \div 50 = 4 \times 2$

$= 8$

g **To divide by 25**, divide by 100 and multiply the answer by 4.

$600 \div 25 = 6 \times 4$

$= 24$

h **To divide by 15**, divide by 30 and double the answer.

$240 \div 15 = 8 \times 2$

$= 16$

2 Now evaluate each expression.

a	32×5	**b**	14×5	**c**	48×5	**d**	18×50
e	52×50	**f**	36×25	**g**	28×5	**h**	12×25
i	12×15	**j**	22×35	**k**	$90 \div 5$	**l**	$170 \div 5$
m	$230 \div 5$	**n**	$1300 \div 50$	**o**	$900 \div 50$	**p**	$300 \div 25$
q	$1000 \div 25$	**r**	$360 \div 45$	**s**	$210 \div 15$	**t**	$360 \div 15$

6.06 Equations and formulas

PATH

Videos
Equations and formulas 2

Equations and formulas 1

Formulas and equations

Puzzle
Getting the right formula

Example 11

The cost of a trip, C, in dollars, for a ride share company is $C = 5 + 2.4d$, where d is the distance travelled, in kilometres.

a Find the cost of a trip if the distance travelled is 15 km.

b Find the distance travelled if the cost of the trip was $78.20.



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Solution

a When $d = 15$:

$C = 5 + 2.4 \times 15$

$= 41$

The cost was $41.

b When $C = 78.20$:

$78.20 = 5 + 2.4d$

$73.20 = 2.4d$

$d = \frac{73.20}{2.4}$

$= 30.5$

The distance travelled was 30.5 km.

Example 12

The surface area of a sphere is given by the formula $A = 4\pi r^2$, where r is the radius.

Find (correct to one decimal place):

a the surface area of a sphere with radius 2.8 cm

b the radius of a sphere with surface area 40 m^2.

Solution

a When $r = 2.8$:

$A = 4 \times \pi \times 2.8^2$

$= 98.520...$

≈ 98.5

The surface area of the sphere is 98.5 cm^2.

b When $A = 40$:

$40 = 4\pi r^2$

$4\pi r^2 = 40$

$r^2 = \frac{40}{4\pi}$

$= 3.183...$

$r = \sqrt{3.183}$ $r > 0$ because the radius is positive.

$= 1.784...$

≈ 1.8

The radius of the sphere is 1.8 m.

PATH

Video Equations and formulas 2

6.06

EXERCISE 6.06 ANSWERS ON P. 631

Equations and formulas UFRC

1 A formula for converting speed expressed in m/s (metres/second) to a speed expressed in km/h is $k = 3.6M$, where M is the speed in m/s. Convert each speed to km/h. R C

EXAMPLE 11

a 10 m/s **b** 24 m/s **c** 50 m/s

2 A car is travelling at a speed of 110 km/h on a freeway. Use the formula from question **1** to calculate how fast this is in m/s.

3 The formula for converting temperatures in °F to °C is $C = \frac{5}{9}(F - 32)$.

Express each temperature in °C, correct to the nearest degree.

a 80°F **b** 32°F **c** 212°F **d** 102°F

4 Use the formula from Question 4 to express each temperature in °F.

a 30°C **b** 40°C **c** 15°C **d** 45°C

5 The average of 2 numbers, m and n, is $A = \frac{m+n}{2}$. If 2 numbers have an average of 28 and one of the numbers is 13, use the formula to find the other number. R

6 Pythagoras' theorem is $c^2 = a^2 + b^2$, where c is the length of the hypotenuse in a right-angled triangle and a and b are the lengths of the other 2 sides. Find, correct to one decimal place where necessary:

EXAMPLE 12

a c, if $a = 5$ and $b = 10$ **b** a, if $c = 41$ and $b = 40$ **c** b, if $c = 20$ and $a = 10$

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7 The formula for the circumference of a circle is $C = 2\pi r$, where r is the radius. Find, correct to one decimal place: R C

a the circumference of a circle with radius 2.4 m

b the radius of a circle whose circumference is 200 cm.

8 The body mass index (BMI) of an adult is $B = \frac{M}{h^2}$, where M is the mass in kilograms and h is the height in metres. Find, correct to one decimal place: R

a the BMI of Leonie who is 1.85 m tall and has a mass of 72 kg

b the mass of Dane with a BMI of 24, who is 2.1 m tall.

9 The volume of a sphere is $V = \frac{4}{3}\pi r^3$, where r is the radius. Find, correct to one decimal place: R

a the volume of a sphere with radius 3.2 cm

b the radius of a sphere with volume 500 m^3.

10 The average speed in km/h of a car is given by the formula $S = \frac{D}{T}$, where D is the distance covered in kilometres and T is the time taken in hours. Find, correct to the nearest whole number: R

a the average speed of a car that takes 4.5 hours to travel a distance of 420 km

b the distance travelled, if a car maintains a speed of 87.2 km/h for 5 hours

c the time taken, if a distance of 650 km is covered at a speed of 91 km/h.

11 The cost, C (in dollars), of a hire car is $C = 75 + 2.5d$, where d is the number of kilometres travelled. Calculate: R

a the cost of hiring a car to travel 350 km

b the distance travelled, if the cost is $135.

12 The surface area of a closed cylinder is given by the formula $SA = 2\pi r^2 + 2\pi rh$. Calculate, correct to one decimal place: R

a the surface area of a cylinder with radius 2.1 m and height 3.5 m

b the height of a cylinder with surface area 1255.38 cm^2 and radius 9 cm.

Changing the subject of a formula 6.07

Example 13

Change the subject of the formula:

a $A = \frac{1}{2}bh$ to h **b** $v^2 = u^2 + 2as$ to s **c** $\frac{a+2}{a+10} = k$ to a

Solution

a $A = \frac{1}{2}bh$

$\frac{1}{2}bh = A$ Swapping sides so that h appears on the LHS

$bh = 2A$ Multiplying both sides by 2

$h = \frac{2A}{b}$ Dividing both sides by b to make h the subject of the formula

b $v^2 = u^2 + 2as$

$u^2 + 2as = v^2$ Swapping sides so that s appears on the LHS

$2as = v^2 - u^2$ Subtracting u^2 from both sides

$s = \frac{v^2 - u^2}{2a}$ Dividing both sides by $2a$ to make s the subject

c $\frac{a+2}{a+10} = k$

$a + 2 = k(a + 10)$ Multiplying both sides by $a + 10$

$= ak + 10k$ Expanding

$a - ak = 10k - 2$ Moving the a-terms to the LHS, the 2 to the RHS

$a(1 - k) = 10k - 2$ Factorising a from the LHS

$a = \frac{10k - 2}{1 - k}$ Dividing both sides by $1 - k$ to make a the subject

PATH

Video Changing the subject of a formula

Worksheet Changing the subject of a formula

Puzzle Formulas squaresaw

PATH

EXERCISE 6.07 ANSWERS ON P. 631

Changing the subject of a formula U F R C

EXAMPLE 13

1 Make y the subject of each formula. R C

a $x + 2y = 5$ **b** $m + py = k$ **c** $P - ky = 8$

d $\frac{m}{3} = \frac{y}{5}$ **e** $D = K - My$ **f** $\frac{5+8y}{d} = 4$

g $\frac{ay-k}{2} = c$ **h** $\frac{y+3}{5} = \frac{4m}{3}$ **i** $xy^2 + 5 = w$

j $x = \sqrt{\frac{y}{k}}$ **k** $n = \frac{d}{5-y}$ **l** $T = \sqrt{\frac{y+k}{c}}$

2 Make h the subject of the formula for the area of a trapezium is $A = \frac{1}{2}(a+b)\,h$. R C

A $h = \frac{A}{2(a+b)}$ **B** $h = \frac{2A}{a} + \frac{2B}{b}$ **C** $h = \frac{2A}{a+b}$ **D** $h = \frac{A}{2a+b}$

3 The formula for the perimeter of a rectangle is $P = 2(L + W)$. Make L the subject.

A $L = P - 2W$ **B** $L = \frac{P-2W}{2}$ **C** $L = W - \frac{P}{2}$ **D** $L = \frac{P}{2W}$

4 The cost ($\$C$) of a taxi trip is $C = 3.6 + 2.2d$, where d is the distance travelled in kilometres. R C

a Make d the subject of the formula.

b If the cost of a taxi trip is \$31.10, find the distance travelled.

5 Change the subject of each formula to the variable indicated in brackets. R C

a $a^2 + b^2 = c^2$ $[b]$ **b** $s = ut + \frac{1}{2}at^2$ $[a]$ **c** $v = u + at$ $[a]$

d $V = \frac{4}{3}\pi r^3$ $[r]$ **e** $A = \pi(R^2 - r^2)$ $[R]$ **f** $A = \pi rl + \pi r^2$ $[l]$

g $S = 180(n - 2)$ $[n]$ **h** $\frac{1}{x} + \frac{1}{r} = \frac{1}{s}$ $[r]$ **i** $x = \sqrt{b^2 - 4ac}$ $[b]$

j $x + y = 5 - 3x$ $[x]$ **k** $m = \frac{5A}{2A+n}$ $[A]$ **l** $S = \frac{a(p-1)}{p}$ $[p]$

m $X(a + b) = Y(a - b)$ $[a]$ **n** $\frac{5+x}{3x+a} = 2$ $[x]$ **o** $y = \frac{u+bx}{u+ab}$ $[b]$

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Graphing inequalities on a number line 6.08

An **inequality** looks like an equation except that the equals sign (=) is replaced by an inequality symbol >, ≥, < or ≤.

$2x - 7 = 15$ is an **equation**. There is only one value of x that makes it true.

$2x - 7 \leq 15$ is an **inequality**. There is a range of values of x that make it true.

PATH

6.08

Worksheet
Graphing inequalities

Inequality symbols

$>$ 'is greater than'

$<$ 'is less than'

$\geq$ 'is greater than or equal to'

$\leq$ 'is less than or equal to'

For example, '$x \geq 3$' is read 'x is greater than or equal to 3'. It includes 3 and all the numbers above 3, such as 3.01, 4, 10, 20 000. '$x > 3$' is read 'x is greater than 3' and means all the numbers above 3, but **not** 3.

Inequality	In words	Meaning
$x > 3$	x is greater than 3	Values above 3
$x < 3$	x is less than 3	Values below 3
$x \geq 3$	x is greater than or equal to 3	Values above and including 3
$x \leq 3$	x is less than or equal to 3	Values below and including 3

An inequality can be graphed on a **number line**.

Example 14

Graph each inequality on a number line.

a $x \geq 1$ **b** $x < 5$ **c** $x > -3$

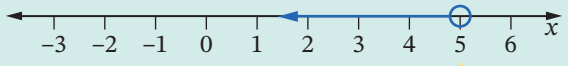
Solution

a $x \geq 1$ means that x can be any number greater than 1 or equal to 1.



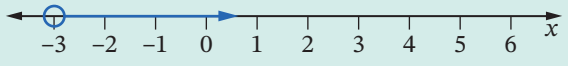
The filled circle at 1 means we include 1.

b $x < 5$ means that x can be any number less than 5, but not including 5.



The open circle on 5 means that 5 is not included.

c $x > -3$ means that x can be any number greater than −3, but not including −3.



Video
Graphing inequalities on the number line

PATH

Example 15

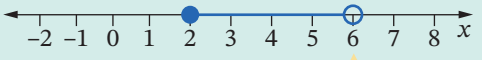
Graph each inequality on a number line.

a $2 \leq x < 6$ **b** $-4 < x < 3$

Solution

a $2 \leq x < 6$ means that x can be any number greater than or equal to 2 but less than 6.

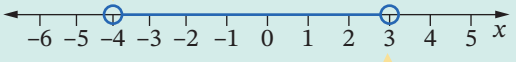
The filled circle at 2 means we include 2.



The open circle at 6 means that 6 is not included.

b $-4 < x < 3$ means that x can be any number greater than -4 but less than 3 (x is between -4 and 3).

The open circle at −4 means that −4 is not included.



The open circle at 3 means that 3 is not included.

EXERCISE 6.08 ANSWERS ON P. 632

Graphing inequalities on a number line U F C

EXAMPLE 14

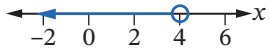
1 Graph each inequality on a separate number line. C

a $x \geq 2$ **b** $x < -3$ **c** $x \leq 1$ **d** $x > 7$

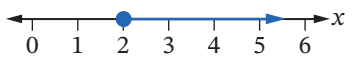
e $x \leq 4$ **f** $x > 0$ **g** $x \geq -2$ **h** $x < 10$

2 Write the inequality shown on each number line. C

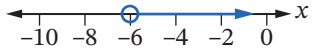
a



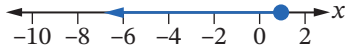
b



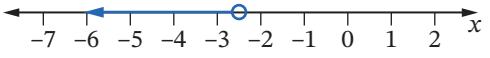
c



d



3 Which inequality is graphed below? Select the correct answer **A**, **B**, **C** or **D**. C



A $x > -2.5$ **B** $x < -2.5$ **C** $x < -3.5$ **D** $x > -3.5$

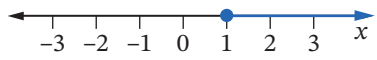
Foundation Standard Complex

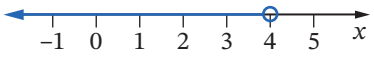
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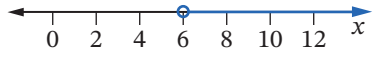
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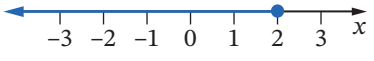
6.08

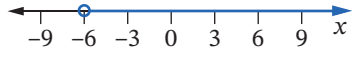
4 Write the inequality shown on each number line. C

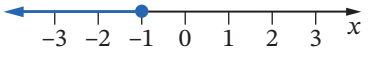
a


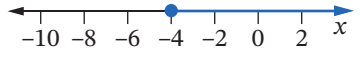
b


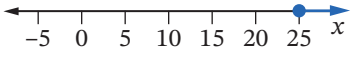
c


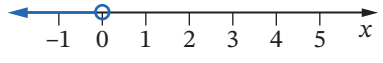
d


e


f


g


h


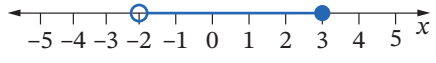
i


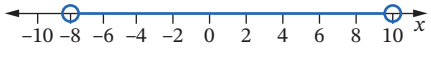
5 Graph each inequality on a separate number line. C

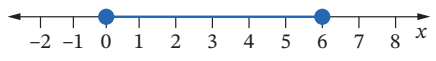
a $0 \leq x \leq 4$ **b** $-2 \leq x < 4$ **c** $-5 < x < 2$ **d** $-1 < x \leq 6$

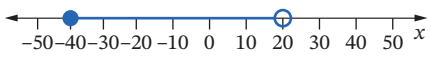
EXAMPLE 15

6 Write the inequality shown on each number line. C

a


b


c


d


Investigation

The language of inequalities

Work in pairs to complete this activity.

Use inequality symbols to write each statement algebraically.

a The minimum height (H) for rides at an amusement park is 1.3 m.

b The speed limit in a school zone is 40 km/h.

c To be eligible to vote, you must be at least 18 years old (A = age).

d The overseas tour is only for people whose age (A) is from 18 to 35.

e The cost (A) of a tennis racquet will be at least \$95, but no more than \$360.

f A new flute (F) costs at least \$475.

g The price of units (U) in a new block start at \$240 000.

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Investigation

Solving inequalities

We have solved equations by doing the same thing to both sides (keeping the equation 'balanced'). Will this method work with inequalities, such as $x + 4 > 10$ or $6x < 13$?

1 Start with an inequality that is true, such as $7 > 4$.

2 Add 5 (or any number you choose) to both sides of the inequality; for example, $7 > 4$ becomes $12 > 9$. Is the new inequality true or false?

3 Subtract 9 (or any number you choose) from each side of the original inequality; for example, $7 > 4$ becomes $-2 > -5$. Is the new inequality true or false?

4 Multiply both sides of the original inequality by 4 (or any positive number you choose); for example, $7 > 4$ becomes $28 > 16$. Is the new inequality true or false?

5 Divide both sides of the original inequality by 2 (or any positive number you choose); for example, $7 > 4$ becomes $3\frac{1}{2} > 2$. Is the new inequality true or false?

6 Multiply both sides of the original inequality by -3 (or any negative number you choose); for example, $7 > 4$ becomes $-21 > -12$. Is the new inequality true or false?

7 Divide both sides of the original inequality by -4 (or any negative number you choose), for example, $7 > 4$ becomes $-1\frac{3}{4} > -1$. Is the new inequality true or false?

8 Which of the 6 operations used in questions **2** to **7** can be used on inequalities to give a true result?

9 Which of the 6 operations used in questions **2** to **7** cannot be used with inequalities because they give a false result?

10 Copy and complete each set of inequality statements.

a $6 < 8$

$6 \times 3 < 8 \times$ ___ (multiplying both sides by 3)

$\therefore 18$ __ 24

b $10 > -4$

$10 \div 2$ __ $-4 \div$ __ (dividing both sides by 2)

$\therefore$ _____

Does the inequality sign ($<$ or $>$) stay the same when multiplying or dividing by a positive number?

11 **a** Is it true that $5 < 8$?

b Multiply both sides by -2. Is it true that $-10 < -16$?

c What needs to be reversed to change $-10 < -16$ into a true inequality statement?

d Copy and complete to make a true statement: -10 _____ -16.

12 **a** Is it true that $18 > -6$?

b Divide both sides by -3. Is it true that $-6 > 2$?

c What needs to be reversed to change $-6 > 2$ into a true inequality statement?

d Copy and complete to make a true inequality statement: -6 ____ 2.

13 Copy and complete: When multiplying or d_____ both sides of an inequality by a n_____ number, the inequality sign must be r_____.

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Solving inequalities

6.09

Example 16

Solve each inequality and graph its solution on a number line.

a $2x - 10 \leq 16$ **b** $2(y - 1) \geq 12$ **c** $\frac{w+3}{2} > -1$

Solution

a

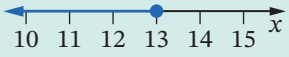
$2x - 10 \leq 16$

$2x - 10 + 10 \leq 16 + 10$

$2x \leq 26$

$\frac{2x}{2} \leq \frac{26}{2}$

$x \leq 13$



Check: Test a value less than 13, let $x = 11$.

LHS $= 2 \times 11 - 10 = 12$

RHS $= 16$

$12 \leq 16$

LHS $\leq$ RHS

b

$2(y - 1) \geq 12$

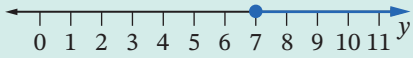
$2y - 2 \geq 12$

$2y - 2 + 2 \geq 12 + 2$

$2y \geq 14$

$\frac{2y}{2} \geq \frac{14}{2}$

$y \geq 7$



Check: Test a value greater than 7, let $y = 8$.

LHS $= 2(8 - 1) = 14$

RHS $= 12$

$14 \geq 12$

LHS $\geq$ RHS

c

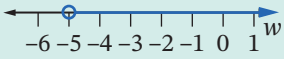
$\frac{w+3}{2} > -1$

$\frac{w+3}{2} \times 2 > -1 \times 2$

$w + 3 > -2$

$w + 3 - 3 > -2 - 3$

$w > -5$



Check: Test a value greater than −5, let $w = -1$.

LHS $= \frac{-1+3}{2} = 1$

RHS $= -1$

$1 > -1$

LHS > RHS

Solving inequalities

- Inequalities can be solved algebraically in the same way as equations, using inverse operations.
- However, when multiplying or dividing both sides of an inequality by a **negative** number, you must **reverse** the inequality sign.

PATH

6.09

Worksheet
Inequalities review

Video
Solving inequalities

PATH

Example 17

Solve each inequality.

a $1 - 2x \geq -11$ **b** $4 - r < 7$ **c** $\frac{a+5}{-3} > 4$

Solution

a
$$1 - 2x \geq -11$$
$$1 - 2x - 1 \geq -11 - 1$$
$$-2x \geq -12$$
$$\frac{-2x}{-2} \leq \frac{-12}{-2}$$
$$x \leq 6$$

Dividing both sides by a negative number reverses the inequality sign

Check: Test a value less than 6, let $x = 5$.

LHS $= 1 - 2 \times 5 = -9$

RHS $= -11$

$-9 \geq -11$

LHS $\geq$ RHS

b
$$4 - r < 7$$
$$4 - r - 4 < 7 - 4$$
$$-r < 3$$
$$\frac{-r}{-1} > \frac{3}{-1}$$
$$r > -3$$

Check: Test a value greater than -3, let $r = 0$.

LHS $= 4 - 0 = 4$

RHS $= 7$

$4 < 7$

LHS $<$ RHS

c
$$\frac{a+5}{-3} > 4$$
$$\frac{a+5}{-3} \times (-3) < 4 \times (-3)$$
$$a + 5 < -12$$
$$a + 5 - 5 < -12 - 5$$
$$a < -17$$

Check: Test a value less than -17, let $a = -20$.

LHS $= \frac{-20+5}{-3} = \frac{-15}{-3} = 5$

RHS $= 4$

$5 > 4$

LHS $>$ RHS

Example 18

Solve each inequality.

a $\frac{3x}{4}+5>2$, where x is a negative integer

b $2-3k \leq 7$, where k is a positive integer

Solution

a
$$\frac{3x}{4}+5>2$$
$$\frac{3x}{4}>-3$$
$$3x>-12$$
$$\frac{3x}{3}>-\frac{12}{3}$$
$$x>-4$$

But x is a negative integer.

$\therefore -4 < x < 0$ for integer x.

$x = -3, -2$ or -1.

b
$$2-3k \leq 7$$
$$-3k \leq 5$$
$$\frac{-3k}{-3} \geq \frac{5}{-3}$$
$$k \geq -\frac{5}{3}$$

But k is a positive integer.

$\therefore k > 0$ for integer k.

$(k = 1, 2, 3, ...)$

EXERCISE 6.09 ANSWERS ON P. 632

Solving inequalities U F R C

1 Solve each inequality and graph its solution on a number line. R C

EXAMPLE 16

a	$x-1>6$	**b**	$3y \geq 12$	**c**	$m+4 \leq 2$
d	$\frac{x}{5} \geq -20$	**e**	$12x < 60$	**f**	$5y > -20$
g	$4a \geq 2$	**h**	$3w \leq -30$	**i**	$8a+5 \geq 45$
j	$3a+1 \leq 10$	**k**	$6a+4 \geq -2$	**l**	$3w-3 < -12$
m	$5a+3 \leq -27$	**n**	$5y+1 \leq 16$	**o**	$4a+5 < 15$

2 Solve $\frac{3(x-2)}{5} \geq -1$. Select the correct answer **A**, **B**, **C** or **D**.

A $x \geq 0$ **B** $x \geq -1$ **C** $x \geq \frac{1}{3}$ **D** $x \geq -\frac{11}{3}$

3 Solve each inequality.

a	$3(x+2) \geq 9$	**b**	$5(m-4) \leq 10$	**c**	$2(y+5) \leq -6$
d	$3+2x < 9+x$	**e**	$11-5y \geq 9-6y$	**f**	$2(3+5a) \leq 5(4+a)$
g	$\frac{x-1}{2} \leq 2$	**h**	$\frac{w-2}{5} > -1$	**i**	$\frac{2a+1}{3} < 3$
j	$\frac{2(m+1)}{3} \leq 3$	**k**	$\frac{4(m-2)}{3} \geq -6$	**l**	$3+\frac{x}{5} < 10$

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EXAMPLE 17

4 Solve each inequality and graph its solution on a number line. R C

a $5 - x \leq 2$ **b** $15 > 7 - y$ **c** $1 - k < 12$

d $7 - m \geq 7$ **e** $2 - p > 8$ **f** $-t + 6 \geq 10$

5 Solve each inequality.

a $-2x < 6$ **b** $\frac{k}{-3} \geq 4$ **c** $-5t > 12$

d $\frac{-x}{3} \leq -4$ **e** $4 - 3w > 7$ **f** $-4y + 3 \leq 11$

g $1 - 3m < 9 - 5m$ **h** $8 - 5a < 3$ **i** $-2d - 3 > 8$

j $\frac{5+w}{-3} > 2$ **k** $3(3x + 4) \geq 6(1 - 2x)$ **l** $\frac{-p+2}{-3} < -2$

EXAMPLE 18

6 Solve each inequality. R C

a $2x + 15 > 3$, where x is a negative integer

b $4 - 3y \leq 16$, where y is a positive integer

c $9 > 1 + 2m$, where m is a positive integer

d $12 \geq 2 - 3w$, where w is a negative integer

Investigation

Power tables

1 Copy and complete this table of powers of 2 from 0 to 20.

x	0	1	2	...	20
2^x					

2 Use the table to calculate 32×128. Explain the method you used.

3 Use the table to calculate:

a 16×1024 **b** 128×2048

c 256×64 **d** 4096×32

4 Use the table to calculate $262\,114 \div 8192$. Explain the method you used.

5 Use the table to calculate:

a $16\,384 \div 512$ **b** $128 \div 8$

c $8192 \div 1024$ **d** $1\,048\,576 \div 65\,536$

When powers are used this way in calculations, they may be called **logarithms.**

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Logarithms 6.10

The **logarithm** of a number is the power of the number, to a given positive base.

For example, the logarithm of 8 to the base 2 is 3, written $\log_2 8 = 3$, because $2^3 = 8$.

PATH



Video
Logarithms



Puzzles
Logarithms 1
Logarithms 2

6.10

Example 19

Evaluate each expression.

a $\log_3 81$ **b** $\log_4 16$ **c** $\log_{10} 10\,000$

Solution

a $\log_3 81$ means $3^? = 81$

Since $3^4 = 81$

then $\log_3 81 = 4$.

'3 to the power of what equals 81?'

b $\log_4 16$ means $4^? = 16$

Since $4^2 = 16$

then $\log_4 16 = 2$.

'4 to the power of what equals 16?'

c $\log_{10} 10\,000$ means $10^? = 10\,000$

Since $10^4 = 10\,000$

then $\log_{10} 10\,000 = 4$.

Logarithms

$$\text{If } y = a^x, \text{ then } \log_a y = x$$

where a is the base, $a > 0$, x is the power, and $y > 0$.

Since $a > 0$, $a^x > 0$ and $y > 0$.

Logarithms are only meaningful for positive numbers, y.

Example 20

Write each expression as a logarithm.

a $243 = 3^5$ **b** $0.01 = 10^{-2}$ **c** $2 = 8^{\frac{1}{3}}$ **d** $p = q^r$

Solution

a $243 = 3^5$

$\therefore \log_3 243 = 5$

b $0.01 = 10^{-2}$

$\therefore \log_{10} 0.01 = -2$

c $2 = 8^{\frac{1}{3}}$

$\therefore \log_8 2 = \frac{1}{3}$

d $p = q^r$

$\therefore \log_q p = r$

PATH

Example 21

Rewrite $\log_n m = x$ in index form.

Solution

$\log_n m = x$ n is the base, x is the power

$\therefore m = n^x$

EXERCISE 6.10 ANSWERS ON P. 632

Logarithms U F R C

EXAMPLE 19

1 Evaluate each expression. R

a $\log_5 25$ **b** $\log_2 8$ **c** $\log_7 49$ **d** $\log_2 16$

e $\log_3 243$ **f** $\log_{10} 1000$ **g** $\log_5 125$ **h** $\log_6 36$

i $\log_2 64$ **j** $\log_3 6561$ **k** $\log_{10} 1\,000\,000$ **l** $\log_8 512$

EXAMPLE 20

2 Write each expression in logarithmic form. R C

a $5^2 = 25$ **b** $4^3 = 64$ **c** $10\,000 = 10^4$ **d** $25^{\frac{1}{2}} = 5$

e $\frac{1}{16} = 2^{-4}$ **f** $3^{-2} = \frac{1}{9}$ **g** $8^{\frac{2}{3}} = 4$ **h** $0.01 = 10^{-2}$

i $\sqrt{2} = 4^{\frac{1}{4}}$ **j** $16^{\frac{1}{2}} = 4$ **k** $9^{\frac{3}{2}} = 27$ **l** $\frac{1}{\sqrt{6}} = 6^{-\frac{1}{2}}$

EXAMPLE 21

3 Write each expression in index form. R C

a $\log_5 125 = 3$ **b** $\log_{10} 10 = 1$ **c** $\log_{\sqrt{3}} 27 = 6$ **d** $\log_2 8\sqrt{2} = 3.5$

e $\log_2 64 = 6$ **f** $\log_3\left(\frac{1}{81}\right) = -4$ **g** $\log_5\left(\frac{1}{125}\right) = -3$ **h** $\log_8 \sqrt{2} = \frac{1}{6}$

i $\log_{100} 10 = \frac{1}{2}$ **j** $\log_5 5\sqrt{5} = \frac{3}{2}$ **k** $\log_8 2 = \frac{1}{3}$ **l** $\log_{100}\left(\frac{1}{100}\right) = -1$

4 Why can't you find the logarithm of a negative number or zero? R C

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Logarithm laws

6.11

The index laws from Chapter 4, *Products, factors and surds*, are related to the logarithm laws.

PATH

Logarithm of a product

The logarithm of a product of terms is equal to the sum of the logarithm of each term.

$$\log_a(xy) = \log_a x + \log_a y$$

For example, $\log_2(8 \times 4) = \log_2 8 + \log_2 4$.

This law corresponds to the index law $a^m \times a^n = a^{m+n}$.

Proof:

Let $m = \log_a x$ and $n = \log_a y$.

$\therefore x = a^m$ and $y = a^n$

$xy = a^m \times a^n = a^{m+n}$

$\therefore \log_a(xy) = m + n$

$= \log_a x + \log_a y$

Logarithm of a quotient

The logarithm of a quotient of terms is equal to the difference between the logarithm of each term.

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

For example, $\log_3\left(\frac{243}{27}\right) = \log_3 243 - \log_3 27$

This law corresponds to the index law $a^m \div a^n = a^{m-n}$.

Proof:

Let $m = \log_a x$ and $n = \log_a y$.

$\therefore x = a^m$ and $y = a^n$

$\therefore \frac{x}{y} = \frac{a^m}{a^n} = a^{m-n}$

$\therefore \log_a\left(\frac{x}{y}\right) = m - n$

$= \log_a x - \log_a y$

Logarithm of a power

The logarithm of a term raised to a power is equal to the power multiplied by the logarithm of the term.

$$\log_a x^n = n \log_a x$$

For example, $\log_4 8^2 = 2 \log_4 8$

This law corresponds to the index law $(a^m)^n = a^{mn}$.

Proof:

Let $m = \log_a x$

$\therefore x = a^m$

$\therefore x^n = (a^m)^n$

$= a^{mn}$

$\therefore \log_a x^n = mn$

$= (\log_a x) \times n$

$= n \log_a x$

PATH

Video
Logarithm laws

Example 22

Evaluate each expression.

a $\log_5 0.04$

b $\log_2 5 - \log_2 10$

c $2\log_3 6 - \log_3 4$

d $\log_5 10 + \log_5 2 - \log_5 4$

Solution

a $\log_5 0.04 = \log_5\left(\frac{4}{100}\right)$

$= \log_5\left(\frac{1}{25}\right)$

$= \log_5\left(\frac{1}{5^2}\right)$

$= \log_5 5^{-2}$

$= -2$ since $\log_a a^x = x$

b $\log_2 5 - \log_2 10 = \log_2\left(\frac{5}{10}\right)$

$= \log_2\left(\frac{1}{2}\right)$

$= \log_2 2^{-1}$

$= -1$ since $\log_a a^x = x$

c $2\log_3 6 - \log_3 4 = \log_3 6^2 - \log_3 4$

$= \log_3 36 - \log_3 4$

$= \log_3\left(\frac{36}{4}\right)$

$= \log_3 9$

$= 2$

d $\log_5 10 + \log_5 2 - \log_5 4 = \log_5 (10 \times 2) - \log_5 4$

$= \log_5 20 - \log_5 4$

$= \log_5\left(\frac{20}{4}\right)$

$= \log_5 5$

$= 1$

Properties of logarithms

$\log_a a^x = x$

$\log_a a = 1$, because $a^1 = a$

$\log_a 1 = 0$, because $a^0 = 1$

$\log_a\left(\frac{1}{x}\right) = -\log_a x$

Proof:

$\log_a\left(\frac{1}{x}\right) = \log x^{-1}$ Using the law $\log_a x^n = n \log_a x$

$= -\log_a x$

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Example 23

Simplify each expression using the logarithm laws.

a $6\log_a a + \log_a a^4 - \log_a a^9$ **b** $\log_2 x + \log_2 w - 2\log_2 y$ **c** $\dfrac{\log_3 a^3}{5\log_3 a}$

Solution

a $6\log_a a + \log_a a^4 - \log_a a^9 = 6 \times 1 + 4 - 9$

$= 1$

b $\log_2 x + \log_2 w - 2\log_2 y = \log_2 (xw) - \log_2 y^2$

$= \log_2\left(\dfrac{xw}{y^2}\right)$

c $\dfrac{\log_3 a^3}{5\log_3 a} = \dfrac{3\log_3 a}{5\log_3 a}$

$= \dfrac{3}{5}$

Example 24

Given $\log_{10} 7 \approx 0.8451$, use logarithm laws to evaluate each expression.

a $\log_{10} 49$ **b** $\log_{10} 700$ **c** $\log_{10} (0.07)$

Solution

a $\log_{10} 49 = \log_{10} 7^2$

$= 2\log_{10} 7$

$\approx 2 \times 0.8451$

$= 1.6902$

b $\log_{10} 700 = \log_{10} (7 \times 100)$

$= \log_{10} 7 + \log_{10} 100$

$\approx 0.8451 + 2$

$= 2.8451$

c $\log_{10} (0.07) = \log_{10}\left(\dfrac{7}{100}\right)$

$= \log_{10} 7 - \log_{10} 100$

$\approx 0.8451 - 2$

$= -1.1549$

Change of base law

The logarithm of a number to the base a can be written as a quotient of logarithms to a different base b. This is called the **change of base law**:

$$\log_a x = \frac{\log_b x}{\log_b a}$$

PATH

Proof:

Let $y = \log_a x$

$\therefore x = a^y$

$\log_b x = \log_b a^y$ Taking logs to the base b of both sides

$\log_b x = y \log_b a$ Using the logarithm of a power law

$\therefore \dfrac{\log_b x}{\log_b a} = y$

But $y = \log_a x$

$\therefore \log_a x = \dfrac{\log_b x}{\log_b a}$

Common logarithms

Logarithms to the base 10 are called **common logarithms**.

The log key on your calculator calculates common logarithms, $\log_{10} x$.

Example 25

a Use the change of base law to express $\log_3 16$ in terms of logarithms to base 10.

b Hence calculate $\log_3 16$ correct to 2 decimal places.

Solution

a $\log_3 16 = \dfrac{\log_{10} 16}{\log_{10} 3}$

b $\log_3 16 = \dfrac{\log_{10} 16}{\log_{10} 3}$ On a calculator, enter log 16 ÷ log 3 =

$= 2.5237 \ldots$ This means $3^{2.5237 \ldots} = 16$.

≈ 2.52

EXERCISE 6.11 ANSWERS ON P. 633

Logarithm laws U F R C

EXAMPLE 22

1 Evaluate each expression. R

a $\log_2 128$ **b** $\log_{10} 1000$ **c** $\log_8 64$

d $\log_5 \frac{1}{5}$ **e** $\log_2 \sqrt{2}$ **f** $\log_3 \frac{1}{9}$

g $\log_{10} 0.0001$ **h** $\log_2 \frac{1}{16}$ **i** $\log_8 2 + \log_8 4$

j $\log_4 32 - \log_4 2$ **k** $\log_{10} 4 + \log_{10} 25$ **l** $\log_5 200 - \log_5 8$

m $\log_2 18 - 2\log_2 3$ **n** $3\log_{10} 2 + \log_{10} 12.5$ **o** $\log_4 192 - \log_4 3$

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2 Simplify each expression. R

a $\log_x 5 + \log_x 6$ **b** $\log_x 10 - \log_x 2$ **c** $3\log_x 2$

d $2\log_x 4 - \log_x 8$ **e** $\log_x 10 + \log_x 4$ **f** $\frac{1}{2}\log_x 100$

g $-\log_x 4$ **h** $\log_x 8 - (\log_x 10 + \log_x 4)$ **i** $\frac{1}{2}(\log_x 8 + \log_x 18)$

3 If $\log_{10} 4 = 0.6021$, evaluate each expression. R

a $\log_{10} 16$ **b** $\log_{10} 400$ **c** $\log_{10} 4000$ **d** $\log_{10} 2$

e $\log_{10} 0.4$ **f** $\log_{10} 160$ **g** $\log_{10} 2.5$ **h** $\log_{10} \sqrt{40}$

4 Evaluate each expression. R

a $\log_3 4 + \log_3 15 - \log_3 20$ **b** $\log_3 270 - (\log_3 2 + \log_3 5)$

c $\log_4 20 + (\log_4 32 - \log_4 10)$ **d** $2\log_{10} 25 - \log_{10} 6.25$

e $2\log_{10} 2 - (\log_{10} 5 + \log_{10} 8)$ **f** $\log_{100} 50 - \log_{100} 5$

g $2\log_5 10 + (\log_5 50 - \log_5 40)$ **h** $5\log_8 2 + \frac{1}{2}\log_8 4$

i $\frac{1}{2}\log_4 25 - 2\log_4 \sqrt{20}$ **j** $\frac{1}{3}\log_2 125 - 3\log_2 \sqrt[3]{80}$

5 Use the change of base law to express each logarithm in terms of logarithms of the given base. C

a $\log_5 50$, base 2 **b** $\log_3 17$, base 7 **c** $\log_9 99$, base 4

6 Use the change of base law to express each logarithm in terms of common logarithms and then calculate its value correct to 2 decimal places. R C

a $\log_3 65$ **b** $\log_7 122$ **c** $\log_5 87$

7 Simplify each expression. R

a $\log_a a^2 + 3\log_a a$ **b** $\log_a (a^3)$ **c** $5\log_a a - \log_a a^4$

d $\dfrac{\log_a (x^7)}{\log_a x}$ **e** $\log_a y^3 - 3\log_a y$ **f** $\log_a \sqrt{x} - \log_a \frac{1}{x}$

PATH

EXAMPLE 23

EXAMPLE 24

EXAMPLE 25

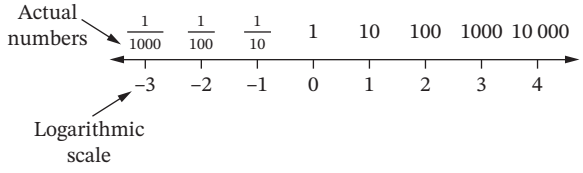
6.12

Logarithmic scales

6.12

Logarithms can be used to describe and graph very small or very large numbers. When it becomes difficult to graph data with very small or large numbers, a **logarithmic scale** is used.

For a base 10 logarithmic scale, the log scale units increase by 1 unit but the actual numbers increase by a power of 10.



PATH

Videos
Logarithmic scales
The Richter scale

Foundation Standard Complex

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Logarithmic scales to base 10 are used for:

- the Richter scale for measuring the magnitude of earthquakes
- the pH scale for measuring acidity of substances
- the decibel scale for measuring loudness

Logarithmic scales that use other bases include:

- the octave scale in music (base 8)
- the F-stops scale for ratios of (light) exposure in photography (base $\sqrt{2}$)

Example 26

What number is represented by each value on a base 10 logarithmic scale?

a 7 **b** −4 **c** 4.85 **d** −0.8

Solution

a Number $= 10^7$ $= 10\,000\,000$

b Number $= 10^{-4}$ $= \frac{1}{10\,000}$ or 0.0001

c Number $= 10^{4.85}$ $\approx 70\,794.578$

d Number $= 10^{-0.8}$ ≈ 0.158

Example 27

How is 157 800 represented on a base 10 log scale?

Solution

Log scale value $= \log_{10} 157\,800$ On a calculator enter: log 157800 =

≈ 5.198

Example 28

Use the base 10 logarithmic scale to determine:

a how many times 1000 is larger than $\frac{1}{100}$

b how much bigger an earthquake measuring 7.3 on the Richter scale is than an earthquake measuring 4.8

Solution

a $\log_{10} 1000 = 3$ and $\log_{10}\left(\frac{1}{100}\right) = -2$

$\log_{10} 1000 - \log_{10}\left(\frac{1}{100}\right) = 3 - (-2) = 5$

$\therefore$ 1000 is $10^5 = 100\,000$ times larger than $\frac{1}{100}$.

b The difference in magnitude $= 7.3 - 4.8 = 2.5$

$\therefore$ The 7.3 earthquake is $10^{2.5} \approx 316.228$ times bigger than a 4.8 earthquake.

EXERCISE 6.12 ANSWERS ON P. 633

Logarithmic scales U F R C

EXAMPLE 26

1 What number is represented by each value on a base 10 logarithmic scale? Answer correct to 3 significant figures for parts **e** to **h**. R C

a 6 **b** −1 **c** 0 **d** 2

e 1.5 **f** −2.3 **g** 4.2 **h** 0.3

EXAMPLE 27

2 How is each number represented on a base 10 log scale? Answer correct to 2 decimal places. R C

a 30 **b** 1 200 000 **c** 0.058 **d** 5000 **e** 0.41

3 By what factor has a value increased if the base 10 logarithmic scale factor has increased by 3? R

EXAMPLE 28

4 Use the base 10 logarithmic scale to determine how many times: R C

a 100 is larger than $\frac{1}{10\,000}$ **b** $\frac{1}{10}$ is smaller than 1 000 000

c an earthquake measuring 8.7 on the Richter scale is bigger than an earthquake measuring 5.9.

5 How much bigger is an earthquake of magnitude 5.6 compared to one of magnitude 4.9? R C

6 The pH formula is given by $pH = -\log_{10}[H^+]$, where $[H^+]$ is the concentration of the H^+ (hydrogen) ions measured in moles/L. Calculate the pH of: R C

a water where $[H^+] = 1 \times 10^{-7}$ moles/L

b orange juice where $[H^+] = 6 \times 10^{-4}$ moles/L

7 **a** For the equation $pH = -\log_{10}[H^+]$, make $[H^+]$ the subject. R C

b If the pH is 5.5, find the value of $[H^+]$.

8 The higher the pH, the lower the acidity of the substance. What is the difference in acidity of 2 soils with pH values of: R C

a 3 and 6? **b** 4.5 and 6.2?

9 The decibel (dB) scale is logarithmic and used to measure 'loudness'. 10 dB increases loudness by a factor of 10 and 20 dB increases loudness by a factor of 100.

By what factor will 30 dB increase loudness? R

10 How much louder is a sound with a loudness of: R C

a 60 dB than a sound of 20 dB? **b** 41 dB than a sound of 38 dB?

c 115 dB than a sound of 70 dB?

PATH

11 This table lists some of the earthquakes experienced in Australia and their strengths on the Richter scale. R C

Year	Location	Strength
1988	Tennant Creek, NT	6.7
1989	Newcastle, NSW	5.6
1997	Collier Bay, WA	6.3
2010	Kalgoorlie, WA	5.2
2011	Bowen-Townsville, QLD	5.4
2015	Coral Sea Fraser Coast, QLD	5.5
2021	Marble Bar, WA	5.3

Use the table to find the difference in magnitude (correct to one decimal place) between the earthquakes in:

a Newcastle and Kalgoorlie

b Collier Bay and Marble Bar

c Tennant Creek and Newcastle

d Bowen-Townsville and Coral Sea

e Collier Bay and Coral Sea

Did you know?

The Richter scale

The Richter scale for the strength of an earthquake uses a base 10 log scale developed by American seismologist and physicist Charles Richter in 1935. As the magnitude of the earthquake increases, the energy released also increases and so too the damage. A large lightning bolt releases as much energy as a magnitude 3 earthquake.

Magnitude	Description
1	Very slight tremor, not felt by anyone.
2	Slight tremor, felt by a few people.
3	Minor earthquake, felt by some people.
4	Light shaking/rattling of objects, felt by most people, minimal damage.
5	Moderate shaking, felt by everyone, furniture moves, slight damage. Kalgoorlie WA (2010): 5.2, Marble Bar WA (2021): 5.3, Bowen-Townsville QLD (2011): 5.4, Newcastle NSW (1989): 5.6.
6	Strong shaking, damage to buildings, felt hundreds of kilometres from epicentre. Collier Bay WA (1997): 6.3, Tennant Creek NT (1988): 6.7, highest in Australia, felt in Perth and Adelaide.
7	Major earthquake causing death and destruction, damage to most buildings within 250 km of epicentre, cracks in the ground, broken pipes underground. Turkiye (Turkey) (2023): 7.8, Japan (2024): 7.6.
8	Major damage to buildings, bridges and structures collapsed, felt across large regions. Peru (2001) and Indonesia (2007): 8.4, Chile (2010): 8.8.
9	Total destruction to all buildings, permanent damage to land, 1-in-10-to-50-year event. Indonesia (2004) and Japan (2010): 9.1.

Earthquakes are rare in Australia. The most destructive earthquake occurred in Newcastle in 1989, causing $4 billion in damage.

Why was Newcastle's earthquake considered the most destructive when Tennant Creek's earthquake in 1988 rated higher on the Richter scale?

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Exponential and logarithmic equations 6.13

Exponential equations are equations like $3^x = 243$, where the variable or unknown is a **power.**

Logarithms can be used to solve exponential equations rather than by a 'guess-and-check' method.

Example 29

Solve each exponential equation.

a $3^x = 243$ **b** $4^{m+1} = \frac{1}{8\sqrt{2}}$

Solution

a

$3^x = 243$

$\log_{10} 3^x = \log_{10} 243$ Taking $\log_{10}$ of both sides

$x \log_{10} 3 = \log_{10} 243$

$x = \frac{\log_{10} 243}{\log_{10} 3} = 5$

On a calculator, enter:
log 243 ÷ log 3 =

Note: The log key means $\log_{10}$, and for convenience we will write log to mean $\log_{10}$.

b

$4^{m+1} = \frac{1}{8\sqrt{2}}$

$\log(4^{m+1}) = \log\left(\frac{1}{8\sqrt{2}}\right)$

$(m+1)\log 4 = \log\left(\frac{1}{8\sqrt{2}}\right)$

$(m+1)\log 4 = -\log(8\sqrt{2})$

$(m+1) = \frac{-\log(8\sqrt{2})}{\log 4}$

$= -1.75$

$m = -2.75$

Example 30

Solve $5^x = 17$, writing the solution correct to 3 decimal places.

Solution

$5^x = 17$

$\log 5^x = \log 17$

$x \log 5 = \log 17$

$x = \frac{\log 17}{\log 5}$

$= 1.7603 \ldots$

≈ 1.760

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6.13

Videos
Exponential equations 2
Exponential equations 1

Worksheet
Logarithms review

Puzzle
Exponential equations

PATH

Logarithmic equations are equations like $\log_5 x = -3$, which can be solved by rewriting the equation in index form.

Example 31

Solve each logarithmic equation.

a $\log_5 x = -3$

b $\log_x 18 = 3$

Solution

a $\log_5 x = -3$

$\therefore x = 5^{-3}$

$= \frac{1}{5^3}$

$= \frac{1}{125}$

b $\log_x 18 = 3$

$\therefore 18 = x^3$

$x = \sqrt[3]{18}$

$= 2.6207...$

≈ 2.62

EXERCISE 6.13 ANSWERS ON P. 633

Exponential and logarithmic equations U F PS R

EXAMPLE 29

1 Solve each exponential equation. R

a $2^k = 512$ **b** $5^m = 78\,125$ **c** $3^d = 59\,049$

d $5^x = 25\sqrt{5}$ **e** $2^y = \frac{1}{16\sqrt{2}}$ **f** $4^a = 128$

g $3^{k+2} = 27\sqrt{3}$ **h** $6^{n-2} = \frac{1}{216\sqrt{6}}$ **i** $9^{1-d} = \frac{1}{27\sqrt{3}}$

EXAMPLE 30

2 Solve each exponential equation, writing the solution correct to 3 decimal places. R

a $7^x = 16$ **b** $5^x = 36$ **c** $11^x = 420$

d $2^x = 0.52$ **e** $3^x = 1.6$ **f** $4^x = \frac{2}{5}$

g $2^{x-2} = 47$ **h** $3^{x+4} = 72$ **i** $6^{x+3} = 29$

j $8^{5-x} = 4000$ **k** $5^y = 4.8$ **l** $7^{k+5} = 300$

3 Solve each exponential equation by expressing both sides to base 2. R

a $2^{x+2} = 16$ **b** $8^x = 32$ **c** $4^{x-1} = \sqrt{2}$

d $8^{1-x} = 16\sqrt{2}$ **e** $4^{2-x} = \frac{1}{8}$ **f** $8^{x+1} = \frac{1}{8\sqrt{2}}$

g $\left(\frac{1}{4}\right)^{1-x} = \sqrt{2}$ **h** $5\left(\frac{1}{2}\right)^x = 20$

EXAMPLE 31

4 Solve each logarithmic equation, correct to 3 decimal places where necessary. R

a $\log_2 x = 3$ **b** $\log_{10} x = 3$ **c** $\log_5 x = -2$

d $\log_4 x = -3$ **e** $\log_{27} x = \frac{1}{3}$ **f** $\log_4 x = \frac{1}{2}$

g $\log_{10} x = -3$ **h** $\log_8 x = \frac{3}{2}$ **i** $\log_{10} x = -\frac{1}{2}$

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j $\log_4 x = -\frac{3}{2}$ **k** $\log_4 x = 3\frac{1}{2}$ **l** $\log_{\sqrt{5}} x = -4$

m $\log_x 4 = 2$ **n** $\log_x 5 = -1$ **o** $\log_x \frac{1}{4} = 2$

p $\log_x 0.01 = 2$ **q** $\log_x 16 = 1$ **r** $\log_x 8 = 3$

s $\log_x 60 = 3$ **t** $\log_x 4.8 = \frac{1}{2}$

5 Use the compound interest formula $A = P(1 + r)^n$ to determine the number of years (to the nearest year) it will take an investment of \$1000 to grow to \$2000, if it earns compound interest at a rate of 6% p.a. PS R

6 Samaria invests \$12 000 at 1% per month compound interest. How many whole months will it take for her investment to grow to \$15 000? PS R

7 A radioactive substance with a mass of 150 grams decays according to the equation $A = 150 \times \left(2^{\frac{-t}{20}}\right)$, where A (grams) is the amount remaining after t days.

Find, correct to the nearest whole number: PS R

a the mass of substance remaining after 10 days

b the time taken for the substance to decay to half its original mass

c the time taken for the substance to decay to a mass of 20 g.

PATH

6.13

Power plus ANSWERS ON P. 633

1 Solve each equation.

a $\frac{2-x}{3} - \frac{x-5}{6} = \frac{5+x}{4} + \frac{x-2}{3}$ **b** $\frac{1}{x-1} + \frac{2}{x+1} = 0$

c $x^2(x-2) - 2(x+1) = x(x^2-3) - 2(5+x^2)$

2 Graph each inequality on a number line.

a $8 \leq x + 7 \leq 11$ **b** $-1 \leq \frac{x}{2} \leq 4$ **c** $-12 < 4x \leq 4$

3 The number of diagonals, D, in a polygon with n sides is $D = \frac{1}{2}n(n-3)$. Show that there is no polygon that has exactly 100 diagonals.

4 The 2 solutions of $x^2 - 8x - 11 = 0$ are in the form $x = p \pm q\sqrt{3}$, where p and q are integers. Find p and q.

5 Solve each logarithmic equation.

a $\log a + \log 3 = \log 21$ **b** $\log x - \log 4 = \log 5$

c $\log 7 + \log m = \log (m + 12)$ **d** $\log (h + 7) - \log 3 = \log (h - 1)$

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CHAPTER 6 REVIEW

Quiz
Language of maths 6

Puzzle
Equations and inequalities crossword

Language of maths

check	cubic equation	exact
expand	exponential equation	factorise
formula	fraction	greater than
inequality	LHS	less than
logarithm	logarithmic equation	logarithmic scale
lowest common multiple (LCM)	number line	quadratic equation
RHS	solution	solve
subject	surd	variable

1 What type of equation has 2 as the highest power of x? Write an example of this type of equation.

2 Write $\log_7 a = -3$ in index form.

3 What is the difference between an **equation** and an **inequality**?

4 Explain why $\log_a\left(\frac{1}{x}\right) = -1$.

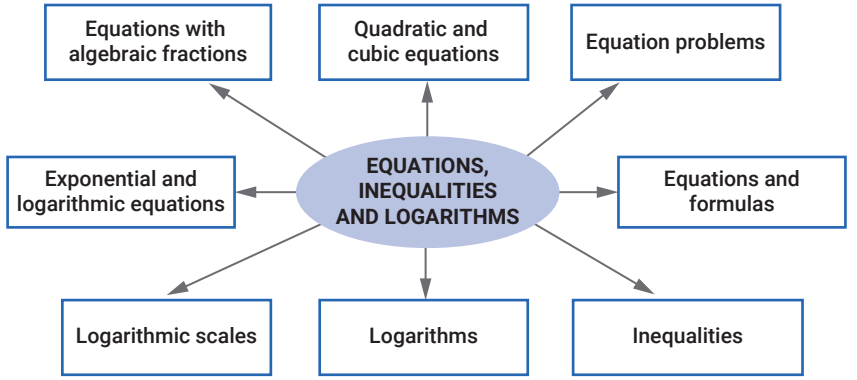
5 When checking the solution to an equation, you need to show that 'LHS = RHS'. What does that mean?

6 What does the inequality symbol '$\leq$' mean?

Worksheet
Mind map: Equations, inequalities and logarithms

Topic summary

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



TEST YOURSELF 6

ANSWERS ON P. 633

Quiz
Test yourself 6

PATH

1 Solve each equation.

a $5k + 17 = 2k + 4$ **b** $20 - 3g = 5g + 6$ **c** $4(3w + 7) = 3(1 - 2w)$

d $7(x + 5) - 2(2 + x) = 24$ **e** $\frac{2a-7}{3} = 4$ **f** $12 + \frac{5m}{3} = 2$

6.01

2 Solve each equation.

a $\frac{3m+5}{6} = \frac{10-m}{3}$ **b** $\frac{2s}{3} - \frac{s}{6} = 2$ **c** $\frac{x}{10} + \frac{x}{2} = 1$

d $\frac{m-1}{4} + \frac{m}{2} = 6$ **e** $\frac{y+1}{3} + \frac{y-1}{4} = \frac{1}{2}$ **f** $\frac{2m-1}{4} - \frac{m-4}{3} = \frac{4}{3}$

6.02

3 Solve each quadratic equation.

a $y^2 = 4$ **b** $p^2 - 100 = 0$ **c** $4x^2 = 40$

d $3m^2 - 3 = 0$ **e** $\frac{2w^2}{5} = 20$ **f** $x^2 + 8x + 7 = 0$

g $h^2 - 8h - 9 = 0$ **h** $u^2 + 4u - 77 = 0$ **i** $k^2 + 5k = 0$

j $m^2 - 2m = 0$ **k** $b^2 + 20b + 100 = 0$ **l** $w^2 = 9w$

6.03

4 Solve each cubic equation, correct to one decimal place.

a $u^3 - 7 = 0$ **b** $5m^3 - 125 = 0$ **c** $\frac{x^3}{2} = 1.5$

6.04

5 Grace is 3 years younger than her sister Jamiela. Twice the sum of their ages is 4 more than their father's age. If their father is 54, find the ages of Grace and Jamiela.

6.05

6 The braking distance (in metres) of a bicycle travelling at a speed of v metres/second is $d = \frac{v(v+1)}{2}$. Calculate the braking distance when the speed of the bicycle is 15 m/s.

6.06

7 The volume of a pyramid is given by the formula $V = \frac{1}{3}Ah$, where A is the area of the base and h is the perpendicular height of the pyramid. Find:

a the volume of a pyramid with a base area of 48 mm^2 and a perpendicular height of 10 mm

b the base area of a pyramid with a volume of 500 m^3 and a perpendicular height of 5 m

6.06

PATH

8 Make a the subject of each formula.

a $y = ax + b$ **b** $P = \sqrt{\frac{a}{m}}$ **c** $M(1 + a) = 1 - a$

6.07

9 Graph each inequality on a number line.

a $x \geq 0$ **b** $x < 3$ **c** $x \leq -2$ **d** $x > -5$

6.08

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6.09 **10** Solve each inequality.

a $y - 6 \geq 10$ **b** $2y \leq -15$ **c** $3a + 10 > -5$

d $10 - 6x < 28$ **e** $\frac{a+2}{-4} > \frac{7}{2}$ **f** $\frac{3-5x}{2} \geq 9$

6.10 **11** Write each expression in index form.

a $\log_6 216 = 3$ **b** $\log_2 \frac{1}{16} = -4$ **c** $\log_7 7\sqrt{7} = \frac{3}{2}$

6.11 **12** Evaluate each expression.

a $\log_7 84 - \log_7 12$ **b** $\log_2 3 + \log_2 \frac{1}{3}$ **c** $2\log_3 9 + \log_3 2 - \log_3 6$

6.11 **13** If $\log_{10} 3 \approx 0.4771$, find the value of:

a $\log_{10} 9$ **b** $\log_{10} 300$ **c** $\log_{10} \frac{10}{3}$ **d** $\log_{10} \sqrt{90}$

6.12 **14** How much bigger is an earthquake measuring 6.3 on the Richter scale than an earthquake measuring 5.9?

6.13 **15** Solve each exponential equation, correct to 3 decimal places.

a $5^x = 11$ **b** $2^x = 0.52$ **c** $3^{x+4} = 105$ **d** $16^{2-x} = 5$